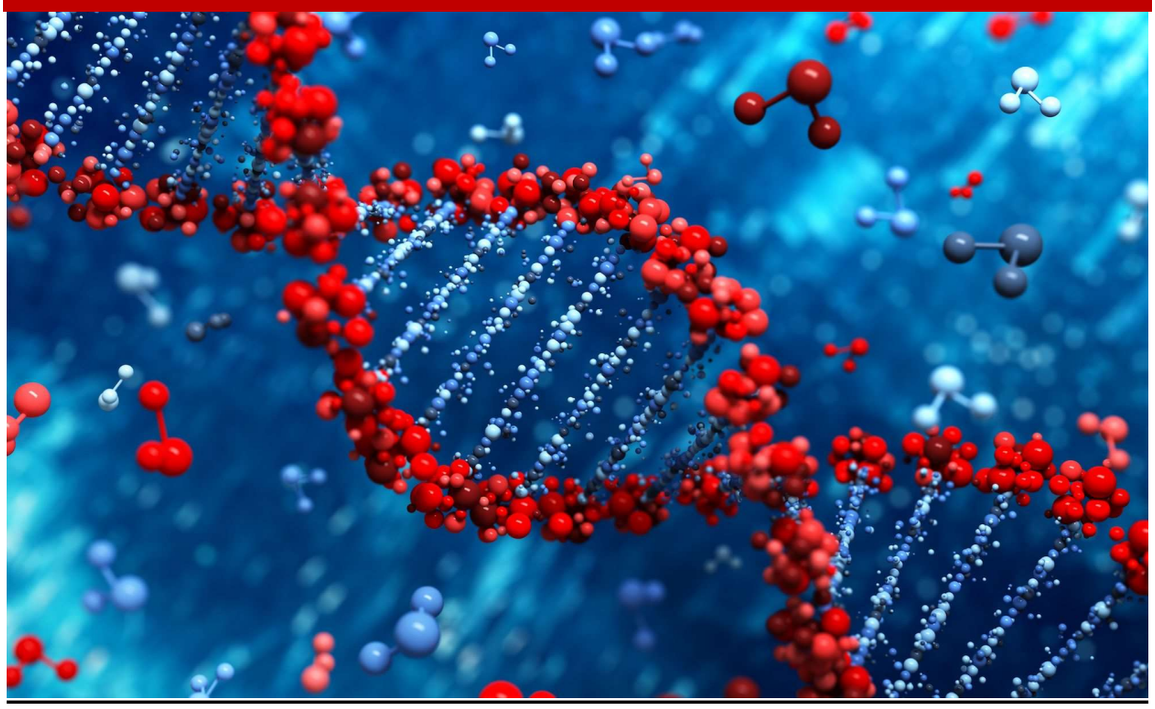
# SUBJECTIVE&OBJECTIVE

# BIO203 - Methods in Molecular Biology

# Final-Term File 2023 By

# SULMAN ALI (BS-ZOOLOGY)

# (0341-7547044)



## CURRENT OBJECTIVES

1. **In situ** Is a polymerase chain reaction that actually takes place inside the cell on a slide.
2. In molecular Technique the term RAPD stands for **Random amplified polymorphic DNA**
3. In Southern Blotting fragments are detected **DNA**
4. Northern blotting is used for **Gene expression**
5. Primary antibody used probes in **Western blotting**
6. How many different SNP screening technique are present. **2**
7. will cut DNA only when the sequence 5'..GAATTC..3' **ECOR1**
8. Enzyme responsible for joining DNA fragments **DNA ligase**
9. **Cohesive** fragment of DNA produced by digestion with EcoR1
10. Enzyme that catalyze template dependent synthesis of DNA **DNA polymerase.**
11. System of cutting DNA molecules was produced by **Smith and Nathan.**
12. Suicide PCR is typically used in **Paleogenetics.**
13. Which of the following Enzyme is responsible for removing phosphate group from many types of molecules including DNA? **Alkaline phosphatase**
14. Which of the following can be used to insert burnt ended foreign DNA into vector? **Ligase**
15. The amount of DNA for RAPD in a reaction is **10ng**
16. Mutation in DNA can be identified by **Southern blot.**
17. PCR _____ **Polymerase chain reaction**
18. RFLP_____ **Restriction fragment length polymorphism**
19. Random Amplified Polymorphic DNA_____**RAPD**

**20.** Amplified fragment length polymorphism_____**AFLP**
**21.** which is type of microscope having beams of electrons_____ **Electron Microscope**
**22.** Which type of pcr is done inside the cell_____**In Situ PCR**
**23.** The type of pcr which defines bacterial yeast clones is..... **Colony PCR**
24. STR stands for.....**Short Tandem Repeats**
25. How many Screening Methods SNPs have.....2
26. WGA PCR stands for..... **Whole Genome Amplification Polymerase Chain Reaction**
**27.** To amplify one strand of the original DNA more than the other, a special type of PCR is used which is known as _____.**Asymmetric Pcr**
28. Which is type of microscope having beams of electrons..... **Electron Microscope**
**29.** Which type of pcr is done inside the cell..... . **In Situ PCR**
**30.** The type of pcr which defines bacterial yeast clones is..... **Colony PCR**
**31.** STR stands for.....**Short Tendem Repeats**
**32.** How many Screening Methods SNPs have.....**2**
**33.** WGA PCR stands for..... **Whole Genome Amplification Polymerase Chain Reaction**
**34.** A type of pcr which amplify single strand..... **Asymmetric Pcr**
**35.** Type of pcr which assembel many pieces into one..... **Assembly Pcr**
**36.** Immunosorbent Electron Microscopy contains..... antibodies. **Both ( monoclonal and polyclonal)**
**37.** In Gel Electrophoresis molecules movement depend on ..... **Size**
38. **11.10%** DMSO
**39.** Works well as a.....**Freezing Medium**
**40.** Antibodies recognize a unique molecule called..... **Antigen**
**41.** How many % attribute make a men unique. **0.1%**
**42.** Denaturation temperature **94 c**
**43.** Annealing temperature **48 to 62 c**
**44.** Extension temperature **72 c**
**45.** Protein of interest is detected and localized using a specific ..... **Antibody**
**46.** wet transfer is recommended for large protein.....**100 kd**
47. RAPD stands for ....**Random amplified polymorphism DNA**
**48.** southern blotting fragments .... **DNA**
**49.** How many different SNP screening **..... 2**
**50.** Enzyme responsible for .....**DNA Ligase**
**51.** cohesive fragment of DNA produced .....**EcoR1**
**52.** suicide PCR is typicaly used bin ..... **paleo genetics**
**53.** How many% attribute men unique **. 0.1%**
54. **Gel electrophoresis** Molecules movement depend on size.
**55.** Immunosorbent Electron Microscopy contains..... antibodies. **Both (monoclonal and polyclonal)**
56. STR stands **short tandem repeats**

# *CURRENT SUBJECTIVE*

## What are the Limitations of PCR?

1= Contamination risk 2=Primer complexities 3= Primer-binding site complexities 4= Amplifies rare species 5=Detection method

## Nested PCR differ Multiplex PCR:

**Nested PCR:** in this type of PCR the Two pairs (instead of one pair) of PCR primers are used to amplify a fragment. First pair amplify a fragment similar to a standard PCR, and have Very low probability of nonspecific amplification.

**Multiplex PCR:** Multiplex PCR is a variant of PCR which enables simultaneous amplification of many targets of interest in one reaction by using more than one pair of primers. The use of

multiple, unique primer sets within a single PCR reaction to produce amplicons of varying sizes specific to different DNA sequences.

## RAPD PCR procedure:

Random amplified polymorphic DNA (RAPD) is a PCR based technique for identifying genetic variation. It involves the use of a single arbitrary primer in a PCR reaction, resulting in the amplification of many discrete DNA products. It does not require any specific knowledge of the DNA sequence of the target organism.

## Advantages of Adenosine monophosphate deaminase (AMPD)

AMP deaminase (AMPD; EC 3.5.4.6) catalyzes hydrolysis of the amino group from the adenine ring of AMP resulting in production of inosine 5'-monophosphate (IMP) and ammonia. This reaction helps to maintain healthy cellular energetics by removing excess AMP that accumulates in energy depleted cells. Furthermore, AMPD permits the synthesis of guanine nucleotides from the larger adenylate pool. This enzyme competes with cytosolic 5'-nucleotidases (c5NT) for AMP. Adenosine, a product of c5NT is a vasodilator, antagonizes inotropic effects of catecholamines and exerts anti-platelet, anti-inflammatory and immunosuppressive activities.

## Hot start PCR:

This is a technique that reduces nonspecific amplification during the initial set up stages of the PCR. This technique may be performed manually by heating the reaction components upto the melting temperature (e.g, 95°C). Before adding the polymerase, specialized enzyme systems have been developed that inhibit the polymerase's activity at ambient temperature. This is done either by the binding of an antibody or by the presence of covalently bound inhibitors that only dissociate after a high-temperature activation step. DNA Polymerase- Eubacterial type I DNA polymerase, Pfu.

## Differentiate between the each Step of PCR:

**1= Denaturation:** This is the First regular cycling event and consists of heating the reaction to 93°C - 98°C for 20- 45 seconds. It causes melting of DNA template yielding single strands of DNA.

**2= Annealing:** In this The reaction temperature is lowered to 40-65°C for 20-40 seconds.Annealing of the primers to the single stranded DNA template.

**3= Extension:** At this step the DNA polymerase synthesizes a new DNA strand complementary to the DNA template strand by adding dNTP's. The temperature at this step depends on the type of DNA polymerase used. Taq polymerase has its optimum activity temperature at 70- 80°C. Commonly 72°C is used.

## Applications Southern blotting:

Southern blotting technique is used to detect DNA in given sample. DNA finger printing is an example of southern blotting. Used for paternity testing, criminal identification, victim identification. To isolate and identify desire gene of interest. t is mostly used in the identification of viral infection and certain bacterial infections.

## Applications of western blotting:

2= Detecting different isoforms of proteins. 2= Detecting protein–protein interactions. 3= Detecting protein–DNA interactions. 4= Detecting post-translational modification. 5= Detecting proteins' subcellular localization. Western blot is often used in research to separate and identify proteins. In this technique a mixture of proteins is separated based on molecular weight, and thus by type, through gel electrophoresis.

## Colony PCR:

The screening of bacterial or yeast clones for correct ligation or plasmid products, and this type of PCR is called Colony PCR.

## Note on restrictions Fragment length polymorphism(RFLP):

It is a difference in homologous DNA sequences that can be detected by the presence of fragments of different lengths after digestion of the DNA samples in question with specific restriction endonucleases.

RFLP analyzes the length of the strands of the DNA molecules with repeating base pair patterns. The basic technique for detecting RFLPs involves fragmenting a sample of DNA by a restriction enzyme, which can recognize and cut DNA wherever a specific short sequence occurs, in a process known as a restriction digest.

The resulting DNA fragments are then separated by length through a process known as agarose gel electrophoresis, and transferred to a membrane via the Southern blot procedure.

## Difference between southern blotting and northern blotting:

1= The Southern blot is used to detect the presence of a particular piece of DNA in a sample by a molecular probe. 2= Southern Blotting is named after its inventor, the British Biologist Edwin Southern (1975). 3=Identify DNA sequence (gene) of interest. 4= Identified DNA may be a small piece of DNA or a mutation.

1= Northern blot is a method used for detection of a specific RNA sequence in RNA samples. 2= This uses Agarose/formaldehyde gel.

## Applications of Western blot:

1= Analysis Of IgG Fractions purified from human plasma.  Diagnosis of HIV by ELISA, involves the western blotting technique. 2= Western blotting technique is also used to Detect Some Forms Of Lyme Disease. 3= Western blotting technique is used in Definitive Test For BSE, which is commonly know as Mad cow disease. 4= Confirmatory Test For Hepatitis-B involves western blotting technique. 5= Western blotting test is used in the Analysis Of Biomarkers such as hormones, growth factors & cytokines.

## Colony PCR:

The screening of bacterial or yeast clones for correct ligation or plasmid products, and this type of PCR is called Colony PCR.

## In Situ PCR:

In Situ PCR (ISH ) is a polymerase chain reaction that actually takes place inside the cell on a slide. In situ PCR amplification can be performed on fixed tissue or cells.

## Asymmetric PCR:

It is used in some types of sequencing and hybridization probing where having only one of the two complementary strands is ideal. It is also used to generate long primers and single-stranded DNA for incorporating cross-linking analogs into specific sites in a DNA probe.

## Joining DNA molecule by alkaline phosphatase.

An enzyme responsible for removing phosphate groups from many types of molecules including DNA. Treatment of linearized plasmid vector DNA with alkaline phosphatase to remove 5'-terminal phosphate groups will prevent recircularization and plasmid dimer formation.

## Electrophoresis of DNA-3:

Electrophoresis is a separations technique that is based on the mobility of ions in an electric field. Positively charged ions migrate towards a negative electrode and negatively-charged ions migrate toward a positive electrode. Ions have different migration rates depending on their total charge, size, and shape, and can therefore be separated.

## Applications of Asymmetric PCR:

It is used in some types of sequencing and hybridization probing where having only one of the two complementary strands is ideal. It is also used to generate long primers and single-stranded DNA for incorporating cross-linking analogs into specific sites in a DNA probe.

## Southern blotting detail:

The Southern blot is used to detect the presence of a particular piece of DNA in a sample by a molecular probe. 2= Southern Blotting is named after its inventor, the British Biologist Edwin Southern (1975). 3= Other blotting methods with similar principles, but using protein or RNA, have been named in reference to Edwin Southern's name. 4= Identify DNA sequence (gene) of interest. 5= Identified DNA may be a small piece of DNA or a mutation.

## Procedure of RFLP:

1= RFLP analyzes the length of the strands of the DNA molecules with repeating base pair patterns. 2= The basic technique for detecting RFLPs involves fragmenting a sample of DNA by a restriction enzyme, which can recognize and cut DNA wherever a specific short sequence occurs, in a process known as a restriction digest. 3= The resulting DNA fragments are then separated by length through a process known as agarose gel electrophoresis, and transferred to a membrane via the Southern blot procedure. 4= Hybridization of the membrane to a labeled DNA probe then determines the length of the fragments which are complementary to the probe. 5= An RFLP occurs when the length of a detected fragment varies between individuals. Each fragment length is considered an allele, and can be used in genetic analysis.

## Full abbreviations of RFLP, RAPD, AFLP:

1=RFLP: Restriction fragment length polymorphism, 2=RAPD: Random Amplified Polymorphic DNA,3= AFLP: Amplified fragment length polymorphism.

## DNA in vitro methods:

1= Joining covalently annealed cohesive ends by DNA ligase 2= Joining blunt-ended fragments by DNA ligase from phage T4 infected E.coli 3= The third utilizes the enzyme terminal deoxynucleotidyl transferase to synthesize homopolymeric3’ single stranded tails at the ends of fragments.

## Negative control and blank reaction

**NEGATIVE CONTROL:** Controls for specificity of the amplification reaction contains all reagents and a DNA template lacking the target sequence.

**BLANK REACTION:** Controls for contamination contains all reagents except DNA template.

## Procedure of DNA profiling:

1= Collect a sample and extract its DNA. Scientists only need a tiny amount of DNA—around 100 micrograms—to construct a DNA profile from a crime scene sample. 2= Amplify the telltale regions. 3= Count the repeats. 4= Look for a match.

## Type 1 of Restriction and Modification (R-M) Type I:

Type I systems were the first to be characterized from E. coli K 12. The active enzyme consists of two restriction subunit, two modification subunit and one recognition subunit. Type I systems are of little value for gene manipulation.

## Define Factors on which proteins migration depends:

Charge, size, and shape.

## Define transfer factors of southern blotting.

1= Upward capillary transfer 2= Downward capillary transfer 3= Simultaneous transfer to two membranes 4= Electrophoretic transfer 5= Vacuum transfer.

## Define the type of pcr in which RNA used as a template:

Northern blot is a method used for detection of a specific RNA sequence in RNA samples. This uses Agarose/formaldehyde gel. This is a capillary transfer. cDNA or RNA probes labeled radioactively or nonradioactive. This is done using an autoradiograph, or detection of light or color change.

## Write a note on ELISA:

Enzyme Linked Immunosorbent Assays

ELISAs are a type of immunoassay that are commonly used to quantify levels of a specific target within a sample. Samples routinely used in ELISAs include serum, plasma, cell culture supernates, cell lysates, saliva, tissue lysates, and urine.

## Different types of polymorphism:

1= Subtype polymorphism (Runtime) 2= Parametric polymorphism (Overloading). 3= Ad hoc polymorphism (Compile-time) 4= Coercion polymorphism (Casting).

## Which type of pcr used in amplification of dna from fossils and how false amplification can be avoided

Multiplex PCR

It can be avoided by using water, tubes, and reagents.

## Define adjuvant:

An adjuvant is an ingredient used in some vaccines that helps create a stronger immune response in people receiving the vaccine. In other words, adjuvants help vaccines work better.

## Types of blotting on the basis of detection:

Different blotting is used to detect different type of macromolecules such as southern blotting is used for DNA analysis, western blotting is for protein analysis, northern blotting is for RNA analysis and eastern for carbohydrate detection.

## Methylation PCR (MSP):

Methylation-specific PCR is used to identify patterns of DNA methylation at CpG islands in genomic DNA. Methylation Specific PCR Methylation Specific PCR METHYLATION SPECIFIC PCR.

MSP is also used in quantitative PCR provides information about methylation state of a given CpG island.

The Target DNA is first treated with sodium bisulfite, which converts unmethylated cytosine bases to uracil, which is complementary to adenosine in PCR primers.

Two amplifications are then carried out on the bisulfite-treated DNA: One primer set anneals to DNA with cytosines (corresponding to methylated cytosine), and the other set anneals to DNA with uracil (corresponding to unmethylated cytosine).

## Explain Immunosorbent Electron Microscope:

The principle of immunosorbent electron microscopy (ISEM) is the selective trapping of plant viruses on to electron microscope grids percolated with a specific antiserum. This technique has been described in a number of papers and review articles to which the reader is referred for comprehensive information. ISEM may be combined with antibody coating (often referred to as "decoration"), a procedure whereby virus particles trapped on the microscope grid are exposed to the homologous antiserum, thus becoming visibly covered with antibody molecules. The consensus is that ISEM is highly reliable (there are virtually no false positives), as sensitive as ELISA, fast (results can often be obtained within one or two hours) and operationally simple (it requires tools and reagents readily available in most laboratories).

## What is meant by real-time PCR?

Real-time polymerase chain reaction (real-time PCR) is commonly used to measure gene expression. It is more sensitive than microarrays in detecting small changes in expression but requires more input RNA and is less adaptable to high-throughput studies.

## Standard PCR reagents:

| Name |
|---|
| DNA |
| Primers R-F |
| Taq Polymerase |
| Buffer Taq |
| Mgcl2 |
| dNTPs |
| Water |

## Difference between DNA ligase and Alkaline phosphatase.

DNA ligase is a basic class of enzymes necessary in joining the strands of DNA together by catalyzing the association of the phosphodiester bond. While Alkaline phosphatase (ALP) is an enzyme that's found throughout your body. An enzyme is a type of protein in a cell that acts as a catalyst and allows certain bodily processes to happen.

## General characteristics of SNPs:

Most commonly, SNPs are found in the DNA between genes. They can act as biological markers, helping scientists locate genes that are associated with disease. When SNPs occur within a gene or in a regulatory region near a gene, they may play a more direct role in disease by affecting the gene's function.

## Effects of snp in coding:

The SNPs located in protein-coding and non-coding RNA genes are classified as neutral and functional. The neutral have no effect, while the functional affect different biological processes and continually confer risk for multifactorial diseases.

## Write a note on VNTR:

DNA profiling uses repetitive sequences that are highly variable, called variable number tandem repeats (VNTRs), particularly short tandem repeats (STRs). VNTR loci are very similar between closely related humans.

VNTR are structural regions of the DNA where a short sequence of nucleotides (longer than 3) is repeated a variable number of times in tandem. VNTRs are commonly subdivided into microsatellites (repeat sequences shorter than 5 nucleotides) and minisatellites (repeat sequences larger than 5 nucleotides) and, like triplet repeats, are thought to be due to DNA slippage errors during DNA replication. As the number of repeats is very individually determined (and generally differs between the maternal and paternal copy), analysis of VNTRs is often used in forensic and paternal identity research.

## Why we cannot use page in the DNA molecules:

DNA is a high molecular weight molecule.It need pore size should be high compare to PAGE so we use gel.

## Microscopy uses:

Microscopy is the technical field of using microscopes to view samples & objects that cannot be seen with the unaided eye (objects that are not within the resolution range of the normal eye).

## Forensic applications:

1= DNA finger printing 2= Detection of Mutations: Inherited disorders & carriers Track DNA abnormalities 3= Prenatal diagnosis of genetic disorders. 4= Detection of pathogens Pre-natal diagnosis DNA fingerprinting Gene therapy Mutation screening 5= Drug discovery 6= Classification of organisms Genotyping.

## Techniques to get known polymorphism.

1= Hybridization Techniques 2= Enzyme based Techniques 3= Direct Sequencing 4= Microarray 5= Cleavage / Ligation 6= Electrophoretic mobility assays.

## Define Alu-PCR and its uses:

The use of primers from a commonly repeated segment is called Alu-PCR.

**Uses:** it can help to amplify sequences adjacent (or between) these repeats.

**Application of PCR in microbiology:** Identification of bacteria: 1= Slow growing bacteria 2= Bacterial antibiotic resistance genes. 3= VIRUS: almost all viruses identification and viral load. 4= FUNGUS 5= Aspergillus spp.

## Types of chromatography:

**There are two main types of chromatography:** liquid chromatography (LC) and gas chromatography (GC). Both LC and GC can be used for either preparative or analytical applications.

## Types of chromatography:

Column chromatography, thin-layer chromatography (TLC), paper chromatography, gas chromatography, ion exchange chromatography, gel permeation chromatography, high-pressure liquid chromatography, and affinity chromatography.

## Enzyme used in binding DNA:

DNA ligase is a DNA-joining enzyme.

## Define types of (R-M)

**Type I:** Type I systems were the first to be characterized from E. coli K 12. The active enzyme consists of two restriction subunit, two modification subunit and one recognition subunit.

**Type II:** Most of the useful R- M system is Type II. Type II enzymes recognize a defined sequence and cut within it

**Type III:** Type III enzymes have symmetrical recognition sequences but otherwise resemble type I systems and are of little value.

**Type Iis:** Type IIs systems have similar cofactors and structure to type II but restriction occurs at a distance from recognition site that limits their usefulness.

## Microsatellites and Minisatellites:

**Microsatellites,** also known as Simple Sequence Repea ts (SSRs) or short tandem repeats (STRs), are repeating sequences of 2-6 base pairs of DNA.

**A minisatellites** (also referred as VNTR) is a section o f DNA that consists of a short series of bases 10– 60 base pairs.

## Agarose gel loading dye:

Gel loading dye is typically made at 6X concentration (0.25% bromphenol blue, 0.25% xylene cyanol, 30% glycerol). That provide density.

## Methodology of PCR:

The PCR technique is based on the enzymatic replication of DNA. In PCR, a short segment of DNA is amplified using primer mediated enzymes. DNA Polymerase synthesises new strands of DNA complementary to the template DNA. The DNA polymerase can add a nucleotide to the pre-existing 3'-OH group only.

## DNA fingerprinting:

DNA fingerprinting is a laboratory technique used to determine the probable identity of a person based on the nucleotide sequences of certain regions of human DNA that are unique to individuals.

## Multiplex PCR:

Multiplex PCR is a variant of PCR which enables simultaneous amplification of many targets of interest in one reaction by using more than one pair of primers. The use of multiple, unique primer sets within a single PCR reaction to produce amplicons of varying sizes specific to different DNA sequences.This technique requires two or more probes that can be distinguished from each other and detected simultaneously. There is a range of different probe technologies available, all using fluorophores.

Multiplex PCR is used in life science research, clinical diagnostics, and forensic laboratories. The development of PCR detection systems with simultaneous multi-target detection and advances in probe chemistries have made comparative analyses standard in many areas of research and testing. The most common uses of multiplex PCR include: SNP genotyping, Pathogen detection, GMO (genetically modified organism) detection, Forensic studies, Food analysis

## Applications of PCR:

Testing of genetic disease mutations, Monitoring the gene in gene therapy, Detecting disease-causing genes in the parents.

PCR is a common tool used in medical and biological research labs. It is used in the early stages of processing DNA for sequencing?, for detecting the presence or absence of a gene to help identify pathogens? during infection, and when generating forensic DNA profiles from tiny samples of DNA.

## TRANSFER METHODS OF DNA TO MEMBRANE:

1= Upward capillary transfer 2= Downward capillary transfer 3= Simultaneous transfer to two membranes 4= Electrophoretic transfer 5=Vacuum transfer.

## What Is Forensic Science?

Forensic science is the use of scientific methods or expertise to investigate crimes or examine evidence that might be presented in a court of law.

## Define haplotype:

A haplotype is a group of genes within an organism that was inherited together from a single parent. The word "haplotype" is derived from the word "haploid," which describes cells with only one set of chromosomes, and from the word "genotype," which refers to the genetic makeup of an organism.

## What is the purpose of triple antibody action

Antibodies are proteins produced by the immune system in response to infection. They are an important part of the body's defence system as they work to destroy disease-causing organisms (such as viruses or bacteria) and block them from infecting human cells.

## Types of snps:

The SNPs in the coding region of gene are divided into two types: synonymous and nonsynonymous SNPs. The synonymous SNPs do not change the amino acid sequence of protein or not affect the protein function. The nonsynonymous SNPs are divided into two types: missense and nonsense.

## What is isoelectric focusing gel?

Isoelectric Focusing Electrophoresis. IEF separates amphoteric compounds, such as proteins, with increased resolution in a medium possessing a stable pH gradient.